

CS Topics: Mobile Application Development

Multimedia Information Display Application Project

*** ALL ITEMS ARE SUBJECT TO CHANGE!!!**

Objective: Develop a mobile application as an interactive information display platform that allows users to explore and learn about various topics of interest. The application must maintain information continuity when the device orientation changes and exhibit a visually pleasing, well-organized interface.

Requirements:

1. Topic Selection

- Choose a theme for the app, such as superheroes, books, music albums, athletes, etc.
- Collect or research additional information about the chosen topic to enhance user engagement. Have about 4-5 data points (i.e., player stats, episode/book details) for each object. You will need at least **ten objects** for your project.

2. ListView Implementation

- Utilize a ListView to present a curated set of choices related to the selected topic.
- Implement a custom adapter for the ListView, incorporating at least three widgets in the XML file (layouts excluded).
- Include a TextView for the name of each item in the list.

3. Two Layout Modes

- Portrait Layout:
 - Display ListView, a relevant image, and a concise description of the selected item.
- Landscape Layout:
 - Retain ListView, image, and description; introduce more detailed attributes specific to the selected item (e.g., powers, aliases, enemies).
 - Rearrange widgets in a landscape layout to optimize screen space.

4. Data Persistence:

- Ensure the app retains information seamlessly as the device's orientation changes.

5. Custom Adapter Features

- Enable user interaction within the ListView using a custom adapter. You will need three widgets in total. The ImageView and buttons will be treated as two widgets. You will need one more of your choice (i.e., SeekBar or RatingBar). You may have to do some research to get these working.
- Custom adapter should be contained in its own class, separate from MainActivity.java.

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- Incorporate a mechanism to modify the list, such as a button to add or remove items.
- Implement visual elements that enhance the user experience (e.g., icons, buttons).

6. Object Representation

- Model the items in the list using objects with a specific class, separate from the MainActivity (e.g., Pokemon.java or Superhero.java), containing instance variables for relevant attributes (e.g., powers, aliases, enemies).
- Implement getter methods to retrieve values in the main program.

7. Aesthetics

- Design the app with a clean and visually appealing layout.
- Ensure widgets are evenly spaced, and there are no empty spaces that give the impression of an unfinished interface.
- Remove image backgrounds to prevent clashes with the app's background.

Submission Guidelines:

- Submit the source code along with any necessary project files.
- Include a brief video outlining the structure, functionality, and additional features implemented.

Note: Refer to the examples and rubric on the next page for further guidance.

Deductions:

- Major Teacher Assistance
- Late Work (deduct 10% per CALENDAR Day)
 - If you plan not to be here/you are sick, you still need to keep up with the project as best as possible. **Speak to Ms. Robles about extensions as needed.**

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Rubric

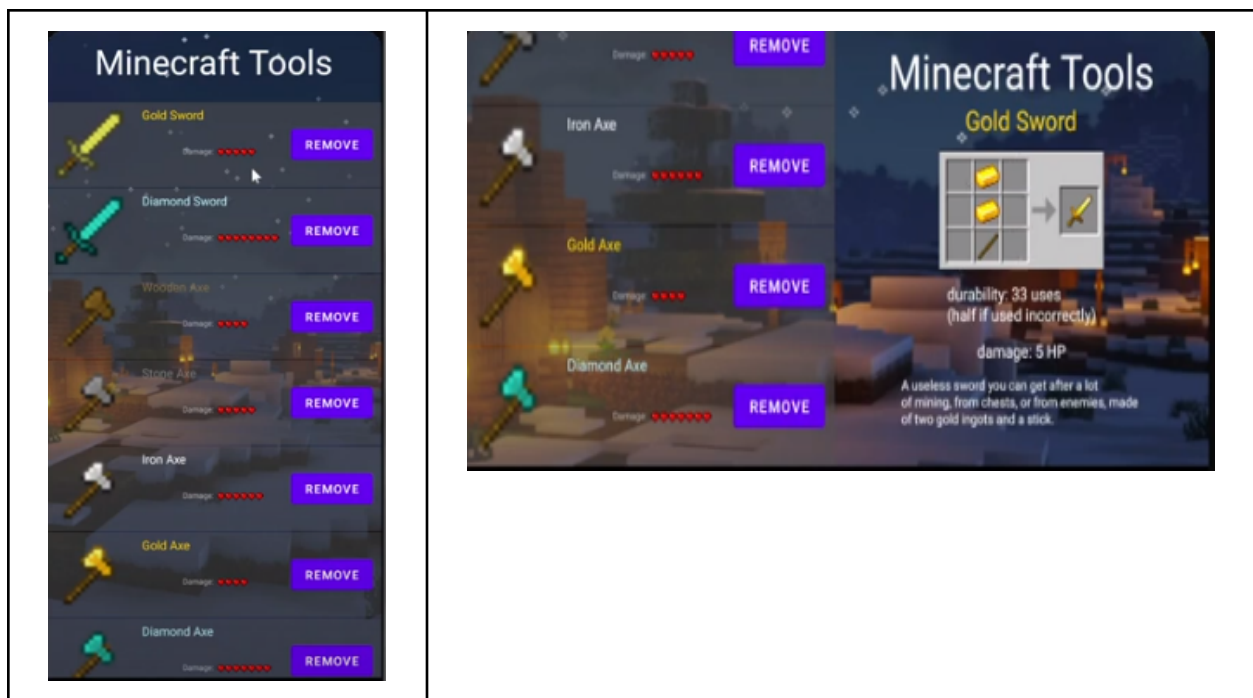
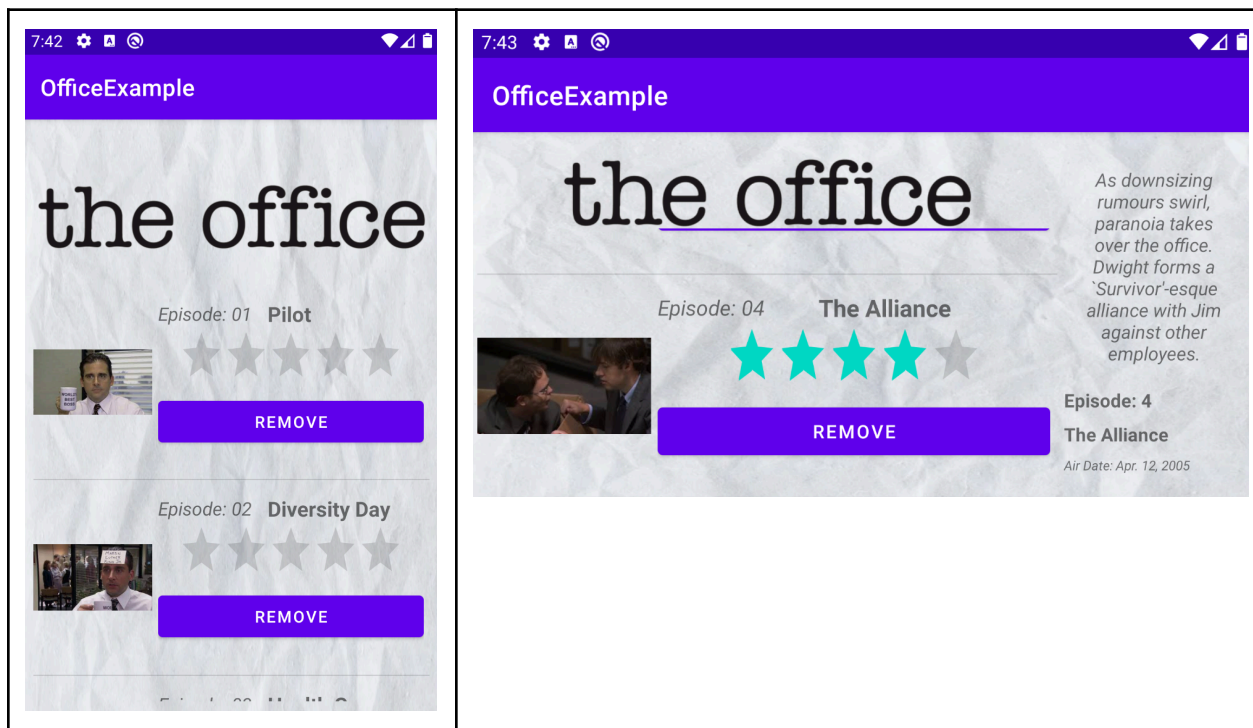
<u>CRITERIA</u>	<u>POINTS</u>
Criteria 1: The app has a basic list (<i>minimum ten items</i>) that shows some information and displays basic information (~4-5 data points) when the user clicks on an item from the list	<u>30 PTS</u>
Criteria 2: The app has an entirely different layout for landscape orientation	<u>30 PTS</u>
Criteria 3: The app displays additional information when the user clicks on an item (see the examples below)	<u>5 PTS</u>
Criteria 4: A custom adapter is used that has different images for each item/type. Custom adapters should be defined in their own class files, not in MainActivity.	<u>5 PTS</u>
Criteria 5: The custom adapter has one (or more) buttons that can alter the original list (this is not shown in all examples below). Typically, this is a remove button.	<u>5 PTS</u>
Criteria 6: The custom adapter employs three <i>different</i> widgets.	<u>5 PTS</u>
Criteria 7: Each item in the list is an Object with a class (the adapter must use this type), and the Objects encapsulate the item's properties. For example, if your item has a year, price, and review, put those variables in the Object, not in MainActivity.java. In the top example below, the Object might be of type OfficeEpisode.	<u>5 PTS</u>
Criteria 9: The list will show the correct items on rotation even after some have been removed (these points will be lost if you cannot remove/add items).	<u>5 PTS</u>
Criteria 10: Layout - Everything should look good on the emulator you have selected. You should remove the background from the images (there should be no white box around the image). Should use a variety of colors/backgrounds and/or text sizes and styles.	<u>5 PTS</u>

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Examples



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